NEUROPLASTICITY

The Brain That Changes Itself

Stories of Personal Triumph from the Frontiers of Brain Science

Norman Doidge, MD · 2007

A psychiatrist tells the stories of the scientists and patients who showed that the adult brain keeps rewiring itself, overturning the old belief that it was fixed for life.[1,2,3]

WHY READ IT

- Gripping true stories of recovery and discovery[3,4]
- An accessible introduction to neuroplasticity[3]
- A hopeful, scientific case that change is possible at any age[4]

GET THE BOOK

The Brain That Changes Itself by Norman Doidge, MD
Viking, 2007. At your local bookshop, your library, or as an audiobook.

The book ↗ Norman Doidge ↗

A free guide inspired by *The Brain That Changes Itself* by Norman Doidge, MD. Not affiliated with the author or publisher.

Please read the book

THE BIG IDEA

What you practise, you become

For a long time, scientists thought each brain region did one fixed job. Doidge tells the story of the research that showed otherwise: what we repeatedly do and pay attention to physically reshapes the brain. That cuts both ways, so choose what you rehearse.[1,3,4]

THREE LESSONS

1 **Use it or lose it**

Brain territory is contested. Skills you practise claim more of it; skills you neglect lose ground.[4,5]

2 **The plastic paradox**

The flexibility that lets us change also locks in rigid habits. Doidge compares it to ski tracks: the more a path is used, the harder it is to leave.[6,7]

3 **Attention is the key**

Doidge reports Michael Merzenich's finding that lasting change in brain maps needs close attention. Divided attention doesn't produce it.[8]

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THE FRAMEWORK

How the brain rewires[3,4]

Four ideas that run through the book's case studies.[3,4]

1 Fire together, wire together

Repetition builds connections

Neurons active at the same time strengthen their links. The phrase sums up Donald Hebb's rule.[10,12]

Try: Pair a habit with a consistent cue and place.

2 Competitive plasticity

Skills compete for space

Practised skills expand; neglected ones shrink.[4,5]

Try: Practise what you want to keep.

3 Mental practice

Imagining changes the brain

Alvaro Pascual-Leone found mental practice of piano changed brain maps much like physical practice.[8]

Try: Rehearse vividly before you practise.

4 Constraint

Block the easy way

Edward Taub's therapy restrains the strong limb so the weak one relearns, in small graded steps.[9]

Try: Remove the shortcut you lean on.

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FROM THE BOOK

Tools to try

Mental rehearsal

Inspired by Pascual-Leone's research in the book.[8]

1. Pick one skill.
2. Rehearse it vividly in your head for five minutes.
3. Then practise it for real with full attention.
4. Track progress over a week.

Interrupt and refocus

From the book's chapter on obsessive loops.[5]

1. When a worry loop or urge starts, name it as a habit circuit, not a fact.
2. Switch to a planned, rewarding activity for about 15 minutes.
3. Log it, and notice the loop weakening over time.

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MAKE IT YOURS

Try it this month

1. Pick one skill to grow.
2. Rehearse it in your head for five minutes a day.
3. Practise for ten minutes with full attention and one shortcut removed.
4. Keep it little and often: steady practice beats cramming.[11]

REFLECT

What have you been unintentionally practising lately?

Which skill are you losing because you stopped using it?

Try it in Sedi: Make the practice a daily Routine in Timely, and keep progress notes in Brainly.

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🧩 The Brain That Changes Itself

Norman Doidge, MD · Viking · 2007

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This guide only scratches the surface. The book has the research, the stories and the nuance behind every idea here. Buy it from an independent bookshop, borrow it from your library, or listen to the audiobook.

A NOTE FROM ME

I'm not affiliated with Norman Doidge, MD or the publisher, and this guide isn't sponsored or endorsed by them. I've learned a lot from this book, so I'm sharing what I took from it, for free, in case it helps you too. Everything here is my own summary, in my own words. The good ideas are the author's; any mistakes are mine.

The maker of Sedi

SOURCES

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